

Vol 5 Chapter 7 — The Born Rule and Measurement

Keeper (author pass — deep math/physics revision)

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Chapter 7 — The Born Rule and Measurement

Legacy chapter (May 2026), corrected in place 2026-09-11 (Keeper, K1892) at the sentences the presentation scan flagged; the remaining tiers are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

What changed since this chapter was written (2026-09-11, Keeper). Three things. (1) The Born rule is derived, from the unique automorphism-invariant measure by Gleason’s theorem (T754); the phrase in that row’s proof, “dimension ≥ 3 because $N_c = 3$,” was decoration — Gleason’s hypothesis is met by any Hardy space — and is removed. (2) The identification of the Born rule with the *Bergman projection* (T2401) stands at version 0.1: its row’s “Bergman exponent 7/2 DERIVED” was a relabel (the exponent is the genus, 5), and no row yet reads a Bergman-normalised state’s Hardy mass as a probability. That link is the one open door of the measurement chain and is named as such in Lecture 3 of the Spine (the Spine, Curriculum/Spine_DIV5_QM_GR_SM/). (3) New since May: on the Šilov boundary the five coordinate write operators satisfy $\sum_u W_u^* W_u = I$ — the write tuple is a resolution of the identity — and Hua’s branching weights $(k+3)/(2k+3)$, $k/(2k+3)$ are its Born probabilities, giving $3/7 = 1/5 + (4/5)(2/7)$ with no table consulted (T2630). Cal’s caveat travels with it: the identity holds on any subset of the unit sphere, so this is a *naming* of the program’s writes as a POVM, not new physics. The four-zone commitment cycle below is the May narrative; its tiers are as stated here, not as stated there.

Level 1 — one sentence

The Born rule $P(n) = |\langle \phi_n | \psi \rangle|^2$ is not a postulate but a *derivation*: the substrate’s Zone 3 commitment step is a Bergman-kernel projection onto an orthonormal outcome basis, and that projection produces exactly $|\langle \phi_n | \psi \rangle|^2$ — Lyra T2401, K67 audit-partial-ready.

Level 2 — graduate-physicist precision

7.1 The Born rule: standard statement

In standard quantum mechanics, measurement is postulated. For a system in normalized state $|\psi\rangle$ and an observable $\hat{A}$ with discrete eigenstates $\{|\phi_n\rangle\}$ and eigenvalues $\{a_n\}$:

- Postulate (measurement): A measurement of $\hat{A}$ yields outcome a_n with probability $P(n) = |\langle\phi_n|\psi\rangle|^2$, and the post-measurement state is $|\phi_n\rangle$ (collapse).
- Postulate (Born rule): $P(n) = |\langle\phi_n|\psi\rangle|^2$.

Born postulated this in 1926; it has been verified to extreme precision but never derived from more fundamental principles within standard quantum mechanics. Various derivations have been attempted (Gleason's theorem, Deutsch-Wallace decision-theoretic, many-worlds branch counting); all rely on additional axioms.

BST claims to derive it from substrate structure alone.

7.2 The 4-zone commitment cycle

Recall from Volume 0 Chapter 3 and Chapter 4 Section 4.2 of this volume: the BST substrate operates in a 4-zone cycle per Koons tick.

- Zone 1 (absorption): external boundary data $\rightarrow$ substrate K-type amplitude
- Zone 2 (bulk computation): unitary K-type evolution under substrate Casimir
- Zone 3 (commitment): Bergman-kernel projection onto outcome basis
- Zone 4 (emission): K-type amplitude $\rightarrow$ external boundary observable

Zones 1, 2, 4 are unitary and preserve the holomorphic structure of $H^2(D_{IV}^5)$. Zone 3 is the non-unitary step — the irreversible projection. This is where the substrate's "measurement" happens.

7.3 Bergman-kernel projection

Setup: substrate is in state $|\psi\rangle \in H^2(D_{IV}^5)$ at the end of Zone 2. The outcome basis $\{|\phi_n\rangle\}$ is the substrate's orthonormal basis for the relevant K-type subspace (e.g., position eigenstates for a position measurement, energy eigenstates for an energy measurement). The Bergman kernel of D_{IV}^5 is $K(z, w)$ — see Chapter 1 Section 1.2.

The substrate's Zone 3 commitment to outcome n proceeds as follows: project $|\psi\rangle$ onto the one-dimensional subspace spanned by $|\phi_n\rangle$:

$$|\psi\rangle \xrightarrow{\text{Zone 3}_n} \frac{|\phi_n\rangle\langle\phi_n|\psi\rangle}{\|\langle\phi_n|\psi\rangle\|}$$

The amplitude $\langle\phi_n|\psi\rangle$ is given by the Bergman inner product:

$$\langle \phi_n | \psi \rangle = \int_{D_{IV}^5} \overline{\phi_n(z)} \psi(z) d\mu(z) = (\phi_n^* \cdot \psi)(K_w) \text{ for some pivot } w$$

The probability of outcome n is determined by the substrate's natural inner product, which inherits the reproducing-kernel structure:

$$P(n) = |\langle \phi_n | \psi \rangle|^2$$

This is the Born rule. Lyra T2401 (Spring 2026, audit-partial-ready) develops the substrate-mechanism derivation in full.

7.4 Why this is a derivation, not a postulate

The standard Born-rule postulate has two parts: 1. Probabilities are the absolute squares of amplitudes 2. The amplitudes are inner products $\langle \phi_n | \psi \rangle$

In BST, both parts come from the substrate:

1. Why squared amplitudes: The Bergman kernel's reproducing property is $f(w) = \langle f, K_w \rangle$. For a normalized state ψ , the substrate's Zone 3 projection onto the one-dimensional subspace spanned by basis element ϕ_n uses the Bergman inner product, and the norm-squared of the projected amplitude is the substrate's natural measure of "weight" — i.e., the probability of that branch. The squaring is unavoidable because the inner product on H^2 is sesquilinear and the projection magnitude is the norm-squared.
2. Why these amplitudes: The substrate's outcome basis $\{|\phi_n\rangle\}$ is the orthonormal basis for the K-type subspace relevant to the observable, and $\langle \phi_n | \psi \rangle$ is the substrate's natural inner product in the Bergman Hilbert space. No additional structure is needed.

The Born rule emerges as the substrate's Zone 3 commitment-mechanism. K67 in the audit chain is the "Born=Bergman" audit; currently audit-partial-ready pending Elie Sessions 6-14 substrate-Hamiltonian closure.

7.5 Worked example: Stern-Gerlach measurement

A spin-1/2 particle in state $|\psi\rangle = a|+\rangle + b|-\rangle$ (with $|a|^2 + |b|^2 = 1$) is sent through a Stern-Gerlach apparatus oriented along $\hat{z}$. The outcome basis is $\{|+\rangle, |-\rangle\}$ — spin up and spin down along $\hat{z}$.

Standard QM: probability of "up" outcome is $|a|^2$; probability of "down" is $|b|^2$. Substrate-mechanism reading: the substrate's Zone 3 commitment projects the K-type onto either the $Pin(2)$ weight $+1/2$ or $-1/2$ component; the projection magnitudes are $|a|$ and $|b|$; the probabilities (norm-squared) are $|a|^2$ and $|b|^2$.

The Stern-Gerlach apparatus is the substrate’s boundary condition for the measurement: external magnetic-field gradient selects which K-type basis the substrate commits to in Zone 3.

7.6 Collapse and the post-measurement state

After substrate commits to outcome n in Zone 3, the post-measurement state is $|\phi_n\rangle$. The “collapse” is the substrate’s irreversible discarding of the orthogonal components (which become decoherent environmental modes, see Chapter 10).

The substrate-mechanism reading: Zone 3 is the substrate’s *time-asymmetric* step. Zones 1, 2, 4 are time-reversible; Zone 3 is not, because the Bergman-kernel projection discards orthogonal amplitude (which leaks into environmental K-types). This is the source of quantum-mechanical irreversibility — the “arrow of time” in measurement.

7.6.5 The multi-tick reading — DCCP candidate principle

Sections 7.3-7.6 treat Zone 3 as a single Bergman-kernel projection per Koons tick. That is correct at the microscopic level: a spin-1/2 measurement against a substrate K-type basis commits in one tick. The substrate doesn’t smear the choice; it makes it.

What happens at the macroscopic level is different — and this is the question Casey’s Discrete Commitment Completion Principle (DCCP) addresses (Casey-named candidate principle, May 24, 2026; current tier FRAMEWORK-PLUS per Cal A. Brate’s #126 referee log).

DCCP claim: at macroscopic scales, where the system is coupled to vast environmental degrees of freedom, Zone 3 commitment runs over many ticks. What standard physics observes as *decoherence* — the gradual loss of quantum coherence as a system entangles with its environment — is the substrate’s commitment phase completing across many K-types simultaneously coupled to environmental modes. The classical world emerges when commitment-in-progress across enough K-types renders quantum-coherent superposition operationally irrelevant.

The arithmetic is concrete. A dust grain of 10^{-15} kg in air at 300 K has a Joos–Zeh decoherence rate of approximately 10^{41} per second. At the Koons-tick rate $t_K \approx 10^{-120}$ s per tick, that is roughly 10^{79} substrate ticks per decoherence event. The dust grain is undergoing a 10^{79} -tick Bergman-commitment cycle. What an observer sees as “the dust grain becoming classical” is the substrate working through that many-tick cycle.

The companion principle, Uncommitted Priors (UP), addresses determinism and agency: once a Zone 3 commitment completes, the outcome is fixed and deterministic. But the chain of pre-commitment moments leading to it — the priors that fed Zone 1 absorption, the substrate states that flowed into Zone 2 computation — remain *uncommitted* until the cycle closes. The substrate has freedom in its priors;

the freedom lives in the chain of uncommitted priors before commitment, not in the commitment itself.

This is the BST-internal reading of the Laplacian determinism question. The substrate is operationally deterministic at the level of individual Zone 3 outcomes (once the projection happens, the result is what it is). But the chain of priors leading to each commitment is open, substrate-modifiable, and contingent on the priors that have not yet committed. *Free will, where it exists, lives in uncommitted priors.*

Tier discipline: DCCP and UP are candidate principles, not ratified. They have framework support, a concrete derivation path (the per-tick commitment scale via the Dirac $Z\alpha = 1$ critical limit gives the predicted decoherence signature step of $\Delta_{DCCP} = 1/N_{\max} \approx 0.730\%$), and a concrete falsifier (the SP-30-1 Bell sub-Tsirelson experimental program testing this signature at 5σ via Vienna IQOQI as of May 24, 2026 outreach). They have not closed at theorem-grade rigor. Volume 14 Chapters 5 and 6 give the current derivation and empirical status.

The reader should carry DCCP forward through the rest of this volume as the multi-tick reading of measurement-induced classicality. When Chapter 10 develops decoherence in detail, it does so on the multi-tick reading. When this chapter says “Zone 3 commits in one tick,” it means at the microscopic scale; at macroscopic scale, the commitment is the many-tick cycle.

7.7 POVMs — generalized measurements

Not every measurement is a projective measurement onto an orthonormal basis. Generalized measurements are described by positive operator-valued measures (POVMs): a collection $\{\hat{E}_n\}$ of positive operators with $\sum_n \hat{E}_n = \hat{I}$ and outcome probabilities $P(n) = \langle \psi | \hat{E}_n | \psi \rangle$.

The standard quantum measurement framework includes POVMs (Chapter 11). In BST: the substrate’s Zone 3 commitment generalizes from Bergman-projection onto an orthonormal basis (projective case) to Bergman-kernel projection onto a more general set of states, with the POVM elements $\hat{E}_n$ inheriting from the substrate’s K-type structure. SP-31-12 extends K67 Born=Bergman to general POVMs; currently pending.

7.8 The measurement problem and quantum interpretations

The “measurement problem” of standard QM asks: what causes the wave function to collapse? Standard answers (Copenhagen, many-worlds, decoherence-as-explanation, hidden variables) each have shortcomings.

BST’s substrate-mechanism resolves the problem by explaining *which substrate process causes collapse*: Zone 3 of the commitment cycle. The “wave function” is the substrate K-type amplitude; “collapse” is Zone 3 projection; “branches” are outcome-amplitude components that the substrate either commits to (selected by Zone 3) or discards (orthogonal complement).

With DCCP (§7.6.5) the picture sharpens further. At the microscopic level, Zone 3 is a single Bergman-kernel projection per tick — clean, discrete, irreversible. At the macroscopic level, Zone 3 commitment smears across many ticks as the substrate works through environmental K-types. What looks to standard physics like “decoherence” is the substrate’s commitment cycle completing across many ticks. The measurement problem dissolves into two specific questions: (1) what substrate-Hamiltonian dynamics drive Zone 3 projection (K52a) and (2) how do many-tick commitment cycles aggregate to produce standard Joos–Zeh decoherence rates (DCCP Lemma 1.4, pending closure).

The substrate-mechanism is not a new interpretation; it is the mechanism that interpretations have been trying to identify. With the mechanism specified, the Born rule is derived (not postulated), the irreversibility is structural (not added), and the measurement problem reduces to specific substrate-dynamics questions with concrete falsifiers.

7.9 K-audit anchors and falsifiers

- T2401 (Lyra Spring 2026): Born=Bergman mathematical derivation
- K67 audit-partial-ready (BST chain): Born=Bergman audit pending Elie Sessions 6-14 substrate-Hamiltonian closure
- SP-31-12 (BST task #283): extend Born=Bergman to general POVMs (Chapter 11 focus)
- 6-audit cascade-unblock pathway includes K67 alongside K52a Lamb shift, K52a BCS, K66 Bell, K68 RS Computation, K69 Universal Q=126

Falsifier: If the substrate-Hamiltonian closure (K52a Sessions 6-14) fails to derive the Bergman-kernel projection by construction from substrate dynamics, K67 Born=Bergman fails, and the substrate-derivation of the Born rule fails. The substrate framework would need either a different mechanism for Zone 3 commitment or to retreat to the Born postulate.

Level 3 — 5th-grader accessibility

When you measure something in quantum mechanics, you don’t get a definite answer until you measure — and the chances of each possible answer are given by squaring the wave function. That squaring used to be a *rule* with no explanation; Born just wrote it down. In BST, the substrate has a 4-step cycle (absorb → compute → commit → emit), and step 3 — “commit” — uses something called the Bergman kernel to pick one answer. When you do the math, the chance of getting answer n is exactly $|\langle\phi_n|\psi\rangle|^2$. The squaring isn’t a rule; it’s what happens when the substrate commits using its built-in inner product. That’s why measurements behave the way they do. It’s like the substrate is asking “how much of this state lines up with that answer?” and the line-up amount, squared, is the chance.

What comes next

Chapter 8 develops Bell's theorem and the CHSH correlations — including BST's signature 1/8 sub-Tsirelson falsifier (SCMP, T2469).

Where to look this up

- Born rule: standard QM textbooks; original Born 1926 paper
- Gleason's theorem: Gleason 1957 (alternative derivation of Born from frame requirement on probability measures)
- BST anchors: T2401, K67 audit, SP-31-12, K52a Sessions 6-14
- DCCP + UP: Casey-named candidate principles, May 24, 2026; FRAMEWORK-PLUS per Cal A. Brate's #126 referee log; full derivation status Volume 14 Chapters 5–6; SP-30-1 Bell experimental program (Vienna IQOQI outreach May 24, 2026)
- Volume 0 Chapter 3 §3.4.5: foundational DCCP framing in substrate operating-system description
- Volume 14 Chapter 5: Born=Bergman from information-theoretic viewpoint + multi-tick DCCP derivation chain
- Joos–Zeh decoherence: Joos, Zeh, et al., *Decoherence and the Appearance of a Classical World in Quantum Theory* (Springer, 2003), cited for the macroscopic decoherence-rate arithmetic in §7.6.5
- Chapter 11 of this volume: POVMs and generalized measurements